



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 10: Producer Theory

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ICS2523 Microeconomics

Lecture 10

Last lecture, we learnt about

- Consumer Welfare
 - Compensated Variation
 - Equivalent Variation
 - Consumer Surplus

In today's lecture, we will start with Producer Theory:

- We will define the production function
- Learn about Marginal and Average Productivity
- Assignment 2



April 2nd Autism Awareness Day

April 2nd **Autism** Awareness Day

- Prevalence: It is estimated that about **1 in 100 children** worldwide has autism and 1 in 36 in the USA.
- **Autism is not an illness.** Available scientific evidence suggests that many factors probably make a child more likely to have autism, including environmental and genetic factors.
- Available **epidemiological data conclude no evidence of a causal association between** the measles, mumps and rubella **vaccine and autism.** Previous studies suggesting a causal link were filled with methodological flaws → [TED Talk by Steve Silberman](#)

April 2nd **Autism** Awareness Day

- Autism spectrum disorders (ASD) are a **diverse** group of conditions → Hence now characterised by a spectrum of colours, not only blue
- They are characterised by some degree of **difficulty with social interaction and communication**.
- Other characteristics are **atypical patterns of activities and behaviours**, such as difficulty transitioning from one activity to another, a **focus on details** and **unusual reactions to sensations**.

April 2nd **Autism** Awareness Day

- **Neurodivergent**: describes people whose brain differences affect how their brain works → they have different strengths and challenges from people whose brains don't have those differences. The possible differences include medical disorders, learning disabilities and other conditions.
- Women and girls with autism spectrum disorder often don't display the behaviours people typically associate with neurodivergence, greatly impacting when, how -- and if -- they are diagnosed → Recommended TED Talk by [Kate Kahle](#),
- Evidence-based psychosocial interventions can improve communication and social skills, positively impacting the well-being and quality of life of autistic people and their caregivers.

The Three Levels of Autism

- **Level 1 - Highest functioning**

Includes those who would have previously been diagnosed with Asperger's syndrome. Individuals with ASD level 1 may have difficulty understanding social cues and may struggle to form and maintain personal relationships. A child with level 1 autism may understand and speak in complete sentences but have difficulty engaging in a back-and-forth conversation.

Children with ASD level 1 experience some inflexibility of behavior, like difficulty switching between tasks, staying organized, and planning.

The Three Levels of Autism

- **Level 2 - Requiring Substantial Support**

Social communication and repetitive behaviours present themselves more obviously in children with ASD level 2 than in children with level 1 autism. Children on this level have challenges in verbal and nonverbal communication and reduced or abnormal responses to social cues.

The inflexibility of behaviour is also more pronounced than in ASD level 1. Repetitive behaviours appear more frequently and may be obvious to casual observers. Likewise, children with level 2 autism may have difficulty coping with changes in routine, which can cause challenging behaviour.

The Three Levels of Autism

- **Level 3 - Requiring Very Substantial Support**

ASD level 3 is characterised by severe challenges in social communication and extremely inflexible behaviour. Children with level 3 autism will be nonverbal or use only a few words of intelligible speech. Initiation of social interaction is very limited, as well as response to others. An individual at this level may interact with others abnormally and only to meet immediate needs.

Individuals with level 3 autism exhibit marked inflexibility of behaviour, with extreme difficulty coping with changes to routine. At this level, restrictive or repetitive behaviours interfere with the individual's ability to function. Changing focus from one activity to another may come with great difficulty and cause significant distress.

Questions from last lecture

2-. A relationship between the Compensated Variation (CV) and Equivalent Variation (EV) is the following:

	CV	EV
When prices rise	+	-
When prices fall	-	+

Announcement

Today there is no lecture at 14:50, I will upload a video with the lecture of how to write an essay.

Producer Theory

1. Profits, Costs and Production

Now we will try to model a firm, so we must understand that:

- Firms make basic choices in their production choices to pursue their goals → usually **profit maximisation**
- We need to understand **costs** to talk about profits
- We need to know something about **production** to understand the costs

<http://www.youtube.com/watch?hl=en&v=rnEFtyMzjOo>

1. Profits, Costs and Production

We will follow these steps:

1. Examine how to model the physical relationship between inputs and outputs → **production function.**
2. We will use the production function to discuss the costs of production → we will assume that all firms seek to **produce their output at the lowest possible cost.**
3. We will investigate the firm's supply decision → firm's manager will make input and output choices to **maximise profits.**

1. Profits, Costs and Production

Is it OK to assume that every firm that maximises profits always makes the right decision?

Everyone makes mistakes, but at the end of the day, in a Market, the firms that will survive will be the efficient ones
→ the non-efficient ones must turn into efficient ones or disappear.

Market definition:

Collection of buyers and sellers that, through their actual or potential interactions, determine the price of a product or set of products

1. Profits, Costs and Production

Now we will focus on how to model the physical relationship between inputs and outputs:

Inputs or Factors of production: Include anything the firm must use as part of the production process. We can divide them into three categories:

- Labour (l)
- Materials (m)
- Capital (k)

Outputs: product

Activity 1

Imagine that you are working in a bakery. Give at least **two examples** for the following:

- a. Labour (input)
- b. Materials (input)
- c. Capital (input)
- d. Product (output)



Código de evento: RUUSQF

<https://app.wooclap.com/RUUSQF?from=instruction-slide>

Production Function

Producer Theory

2. Production function

Principal activity of any firm is to **transform inputs** into **outputs**.

We will not discuss the engineering intricacies involved in the elaboration of the output

→ we will work with an **abstract model of production**

2. Production function

Production function will describe the relationship between the inputs and the resulting output (or service):

$$q = f(k, l, m, \dots)$$

- q represents the firm's output of a particular good during a period
- k represents the capital usage during that period
- l represents hours of labour input
- m represents the raw materials used

Lowercase $q \rightarrow$ One firm's output

Uppercase $Q \rightarrow$ Total output in a market

2. Production function

The production function will allow that inputs to combine in different combinations to produce the output
→ **final output may be produced in different ways**

Production function describes what is **technically feasible** when the firm operates efficiently.

Marginal and Average Productivity

Producer Theory

3. Marginal Physical Product

Marginal Physical Product of an input

Is the **additional output that can be produced** by employing one more unit of that input while *holding all other inputs constant* :

Marginal physical product of capital:

$$MP_k = \frac{\partial q}{\partial k} = f_k$$

Marginal physical product of labour:

$$MP_l = \frac{\partial q}{\partial l} = f_l$$

2. Average Physical Product

Average Physical Product of an input

Is the **total production by unit of that input** while *holding all other inputs constant* :

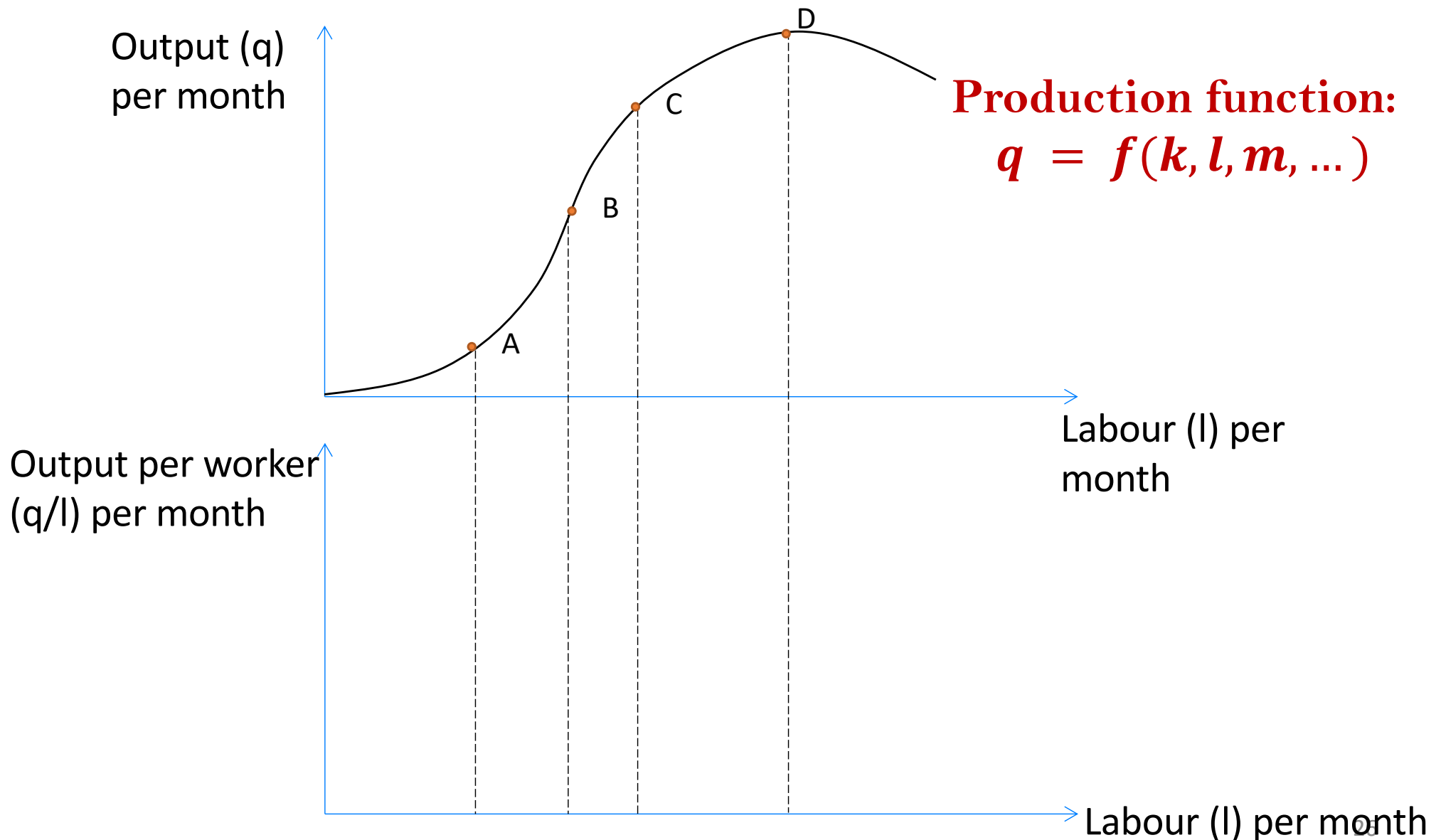
Average physical product of capital:

$$AP_k = \frac{q}{k} = \bar{f}_k$$

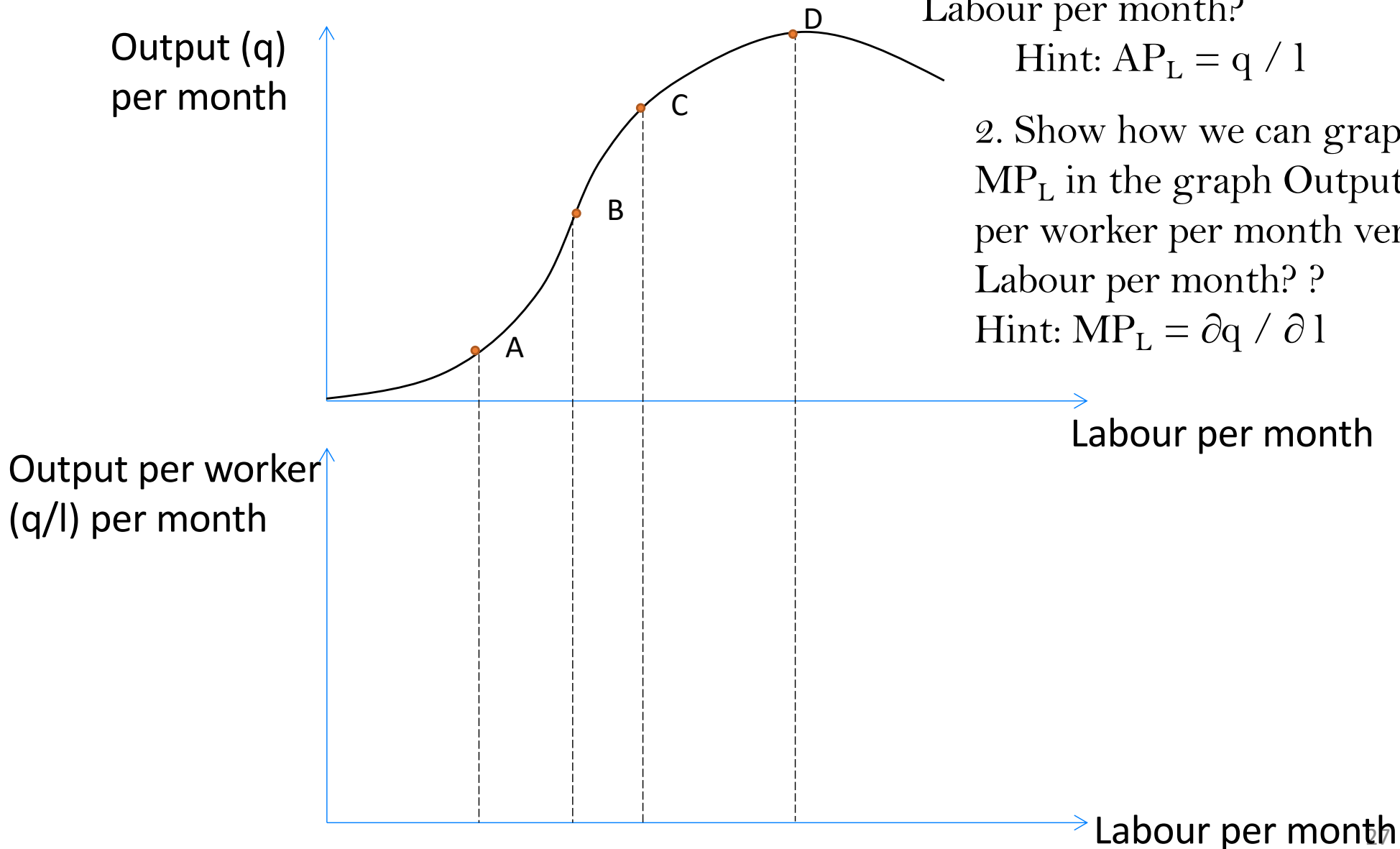
Average physical product of labour:

$$AP_l = \frac{q}{l} = \bar{f}_l$$

The Slopes of the Production Curve



Example



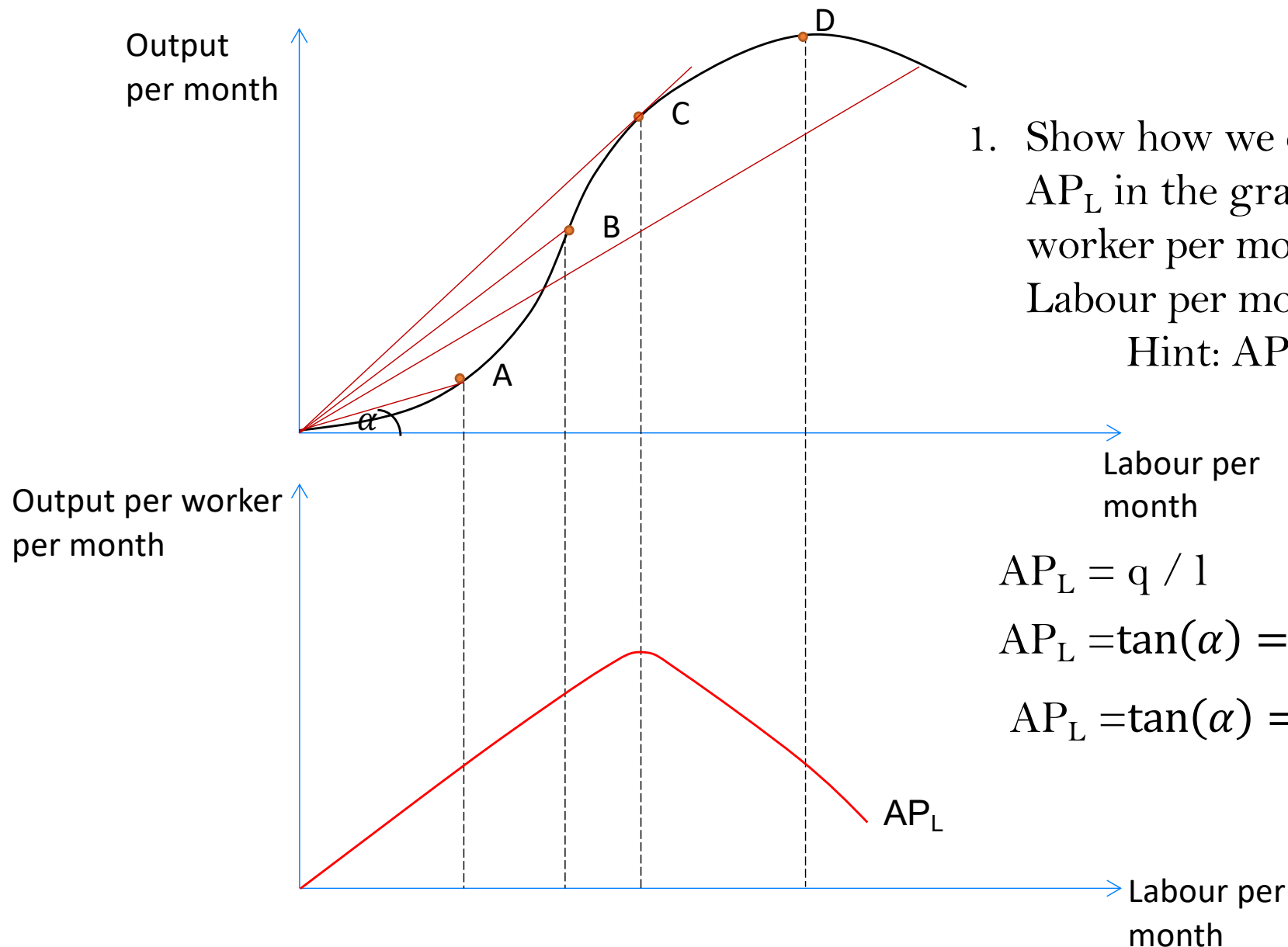
1. Show how we can graph AP_L in the graph Output per worker per month versus Labour per month?

Hint: $AP_L = q / l$

2. Show how we can graph MP_L in the graph Output per worker per month versus Labour per month? ?

Hint: $MP_L = \partial q / \partial l$

Example: Solution



1. Show how we can graph AP_L in the graph Output per worker per month versus Labour per month?

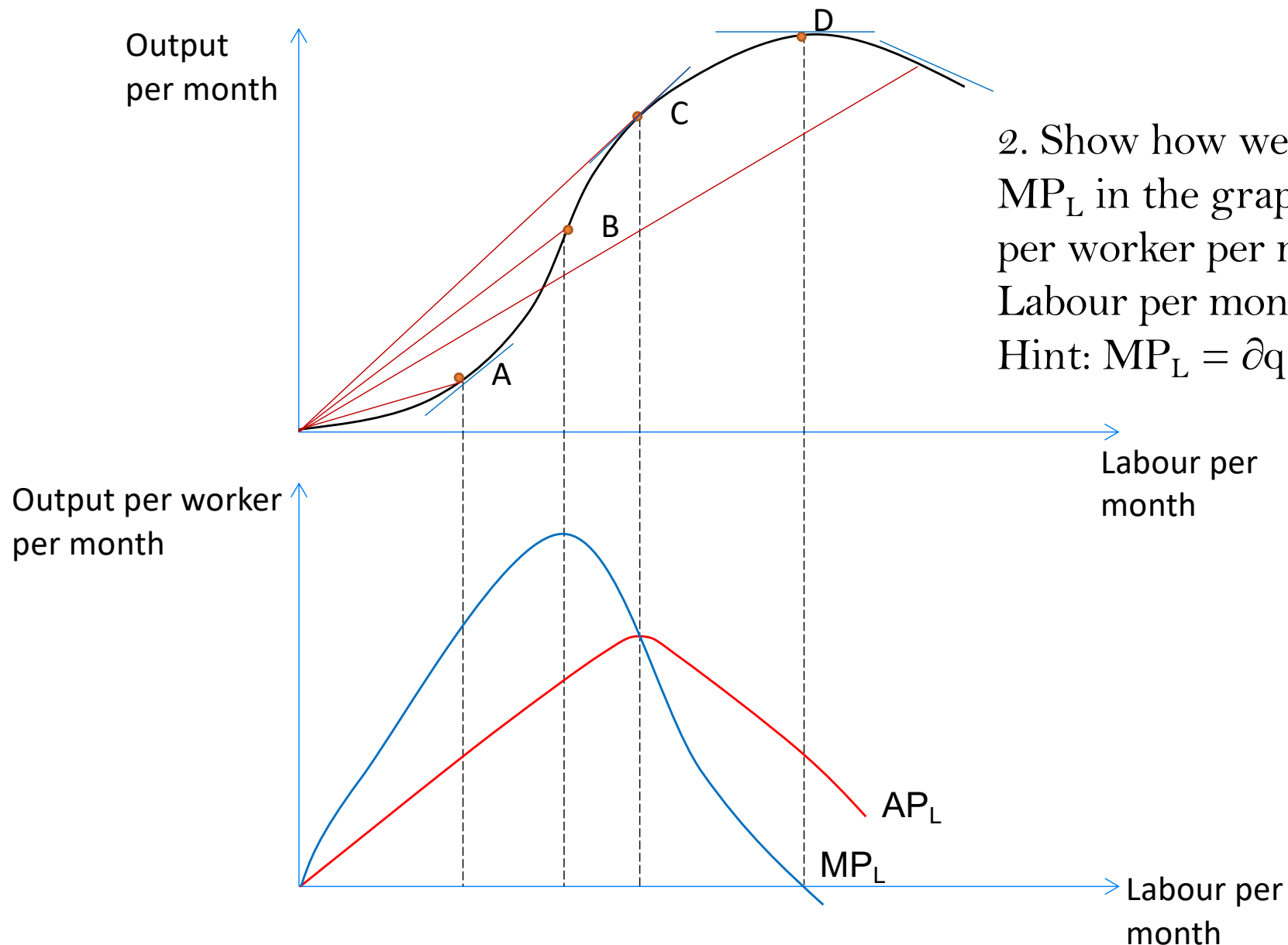
Hint: $AP_L = q / l$

$$AP_L = q / l$$

$$AP_L = \tan(\alpha) = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$AP_L = \tan(\alpha) = \frac{q}{l}$$

Example: Solution



2. Show how we can graph MP_L in the graph Output per worker per month versus Labour per month? ?
Hint: $MP_L = \partial q / \partial l$

Activity 2

3. Under what condition the maximum of AP_L is found?

Hint: derivate AP_L

Law of Diminishing Marginal Returns

The marginal productivity of an input depends on how much of that input is used.

Suppose we are harvesting carrots in a particular field. In that case, labour cannot be added indefinitely to a given field (while keeping the amount of equipment, fertiliser, land, etc. fixed) without degrading its productivity.

This means $\frac{\partial MP_k}{\partial k} = \frac{\partial^2 f}{\partial k^2} < 0$ and $\frac{\partial MP_l}{\partial l} = \frac{\partial^2 f}{\partial l^2} < 0$

This is also called **The Law of Diminishing Marginal Returns**: as the use of an input increases with other inputs fixed, the resulting additions to output will eventually decrease.

Activity 3

If the production function is given by a Linear production function:

$$q = f(k, l) = \alpha k + \beta l$$

- a. Graph the production function versus input k
- b. Find an expression for MP_k
- c. Find an expression for AP_k

Homework L11

If the production function is given by a Cobb-Douglas (where $\alpha + \beta = 1$)

$$q = f(k, l) = Ak^{\alpha}l^{\beta}$$

- a. Graph the production function versus input k
- b. Find an expression for MP_k
- c. Find an expression for AP_k

Isoquants and Rate of Technical Substitution

Producer Theory

4. Isoquants and Rate of Technical Substitution

An **isoquant** shows all the possible combinations of inputs that yield the same quantity of product

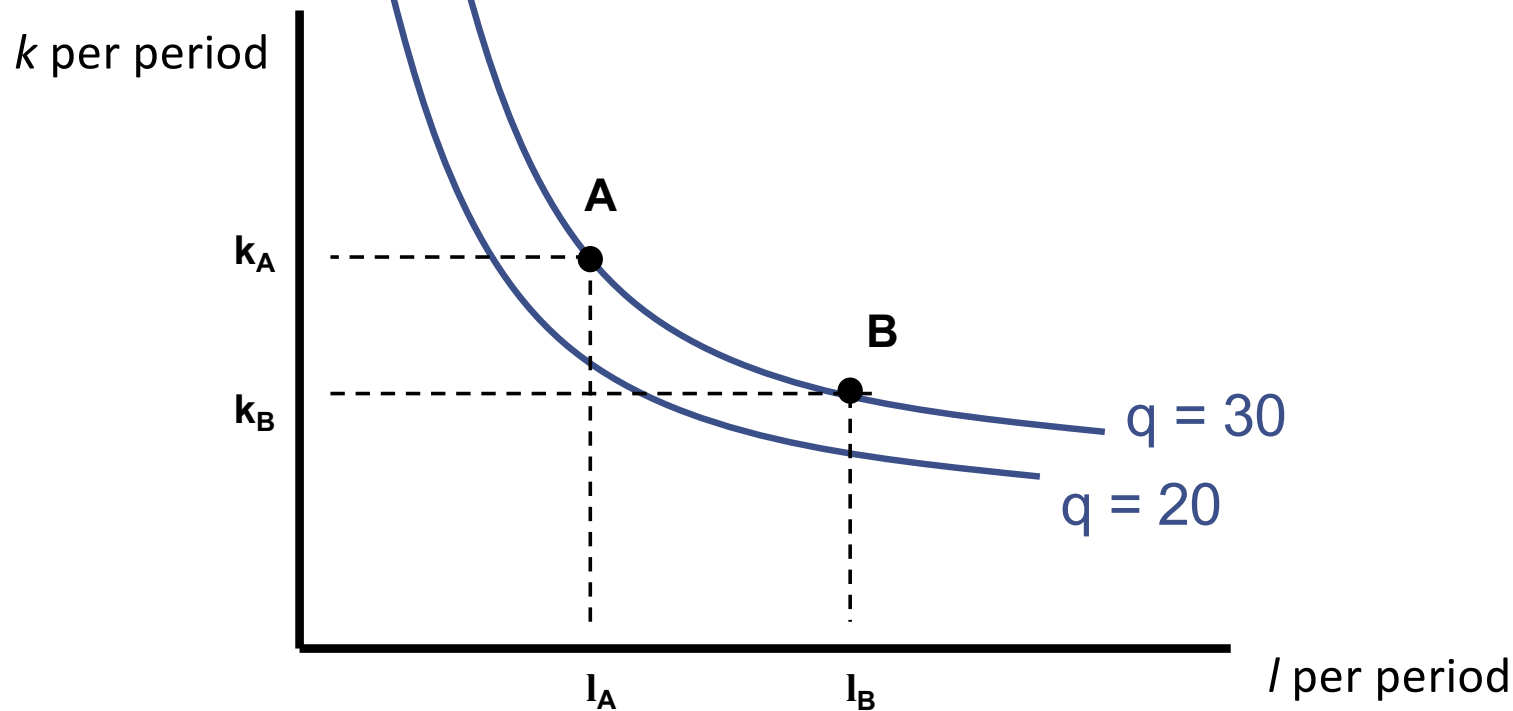
$$f(k, l) = q_o$$

The **slope** of an isoquant illustrates the rate at which l can be substituted for k by holding output constant → **Rate of Technical Substitution (RTS)**

RTS > 0 and is diminishing for increasing inputs of labour

4. Isoquants and Rate of Technical Substitution

$$RTS(l \text{ for } k) = - \left. \frac{dk}{dl} \right|_{q=q_0} = \frac{\partial f(k, l) / \partial l}{\partial f(k, l) / \partial k} = \frac{MP_l}{MP_k}$$



4. Isoquants and Rate of Technical Substitution

Marginal Rate of Technical Substitution (RTS)

$$RTS(l \text{ for } k) = - \left. \frac{dk}{dl} \right|_{q=q_0} = \frac{\partial f(k, l) / \partial l}{\partial f(k, l) / \partial k} = \frac{MP_l}{MP_k}$$

How much I can reduce the quantity of k when an extra unit of l is used, so that output remains constant

4. Isoquants and Rate of Technical Substitution

Marginal Rate of Technical Substitution (RTS)

If we set the total differential of the production function:

$$dq = \frac{\partial f}{\partial l} dl + \frac{\partial f}{\partial k} dk = MP_l dl + MP_k dk$$

Since the production is being held constant ($dq = 0$), rearranging gives that:

$$MP_l dl + MP_k dk = 0$$

$$RTS(l \text{ for } k) = - \left. \frac{dk}{dl} \right|_{q=q_0} = \frac{MP_l}{MP_k}$$

$$RTS > 0$$

Author: Javiera San Martín (2020)

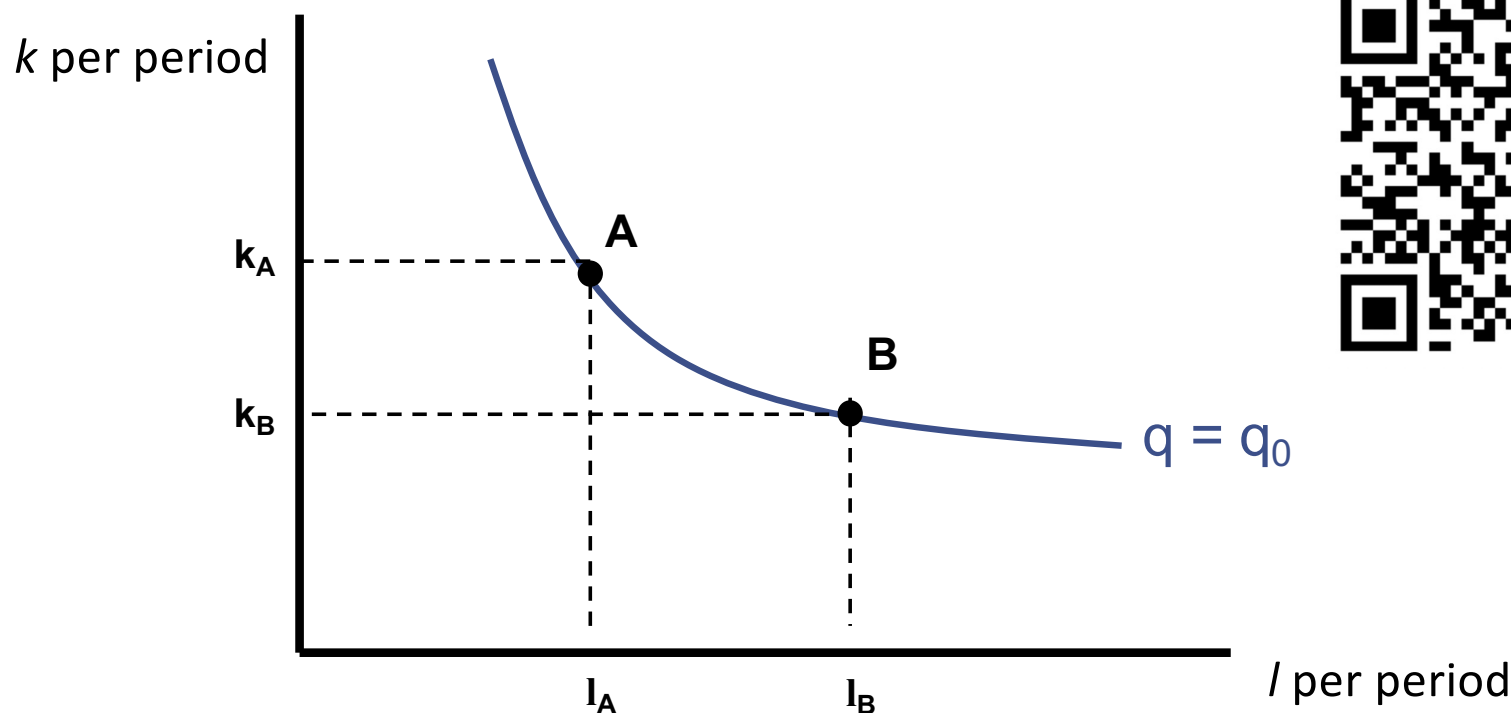


**when in class
you saw
consumer
theory and now
you start to see
producer theory**

Reference: Facebook
(someone sent me this image)

Activity 4

Along the isoquant curve (if we increase the quantity of labour used in production), the RTS increases, is constant or decreases?



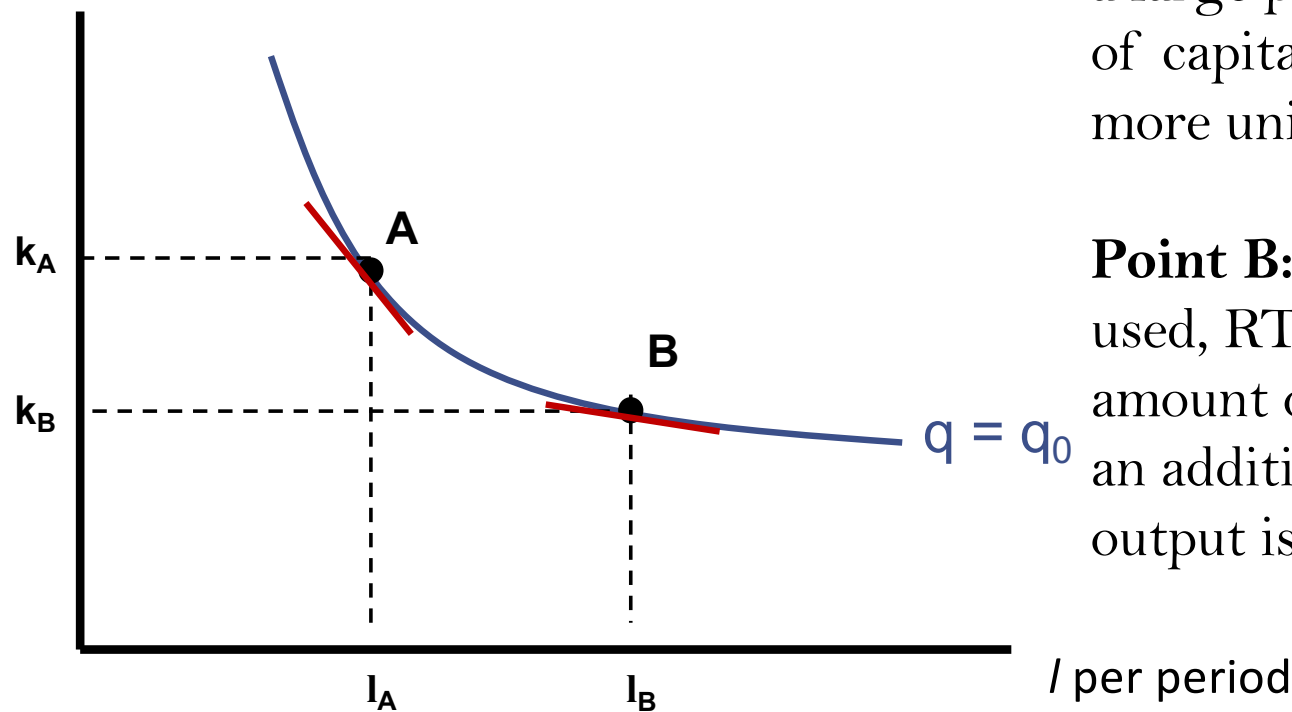
Código de evento: RUUSQF

<https://app.wooclap.com/RUUSQF?from=instruction-slide>

4. Isoquants and Rate of Technical Substitution

Isoquants have **negative slope and convex** →
Along the curve **RTS is diminishing**.

k per period



Point A: high ratio of k to l , RTS is a large positive number → great deal of capital can be given up if one more unit of labour is available

Point B: A lot of labour is being used, RTS is low → only a small amount of capital can be traded for an additional unit of labour (if output is to be held constant)



When you hire more workers to have more hours of labour and rise the production



But the law of diminishing marginal returns applies *ceteris paribus* and say you that the production eventually decrease.

source:

<https://www.infobae.com/teleshow/infoshow/2018/07/01/drake-el-exitoso-rapero-con-cara-de-meme/>

Activity 5

Graph the **isoquants** of the following production functions:

- a) Linear production function

$$q = f(k, l) = \alpha k + \beta l$$

- b) Fixed proportions

$$q = f(k, l) = \text{Min} \{ \alpha k, \beta l \}$$

- c) Cobb-Douglas

$$q = f(k, l) = Ak^{\alpha}l^{\beta}$$

Individual Activity

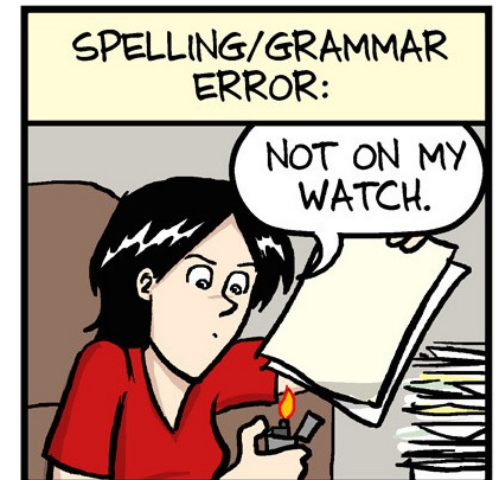
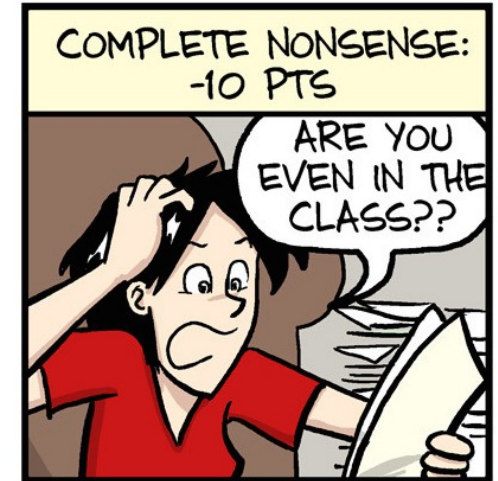
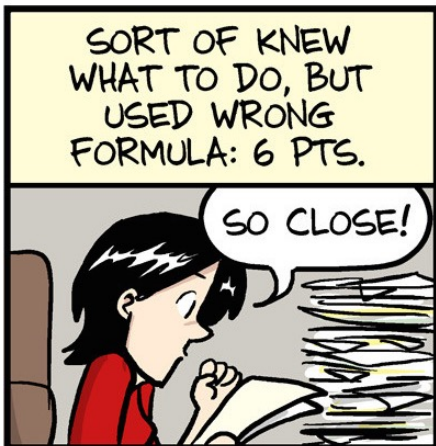
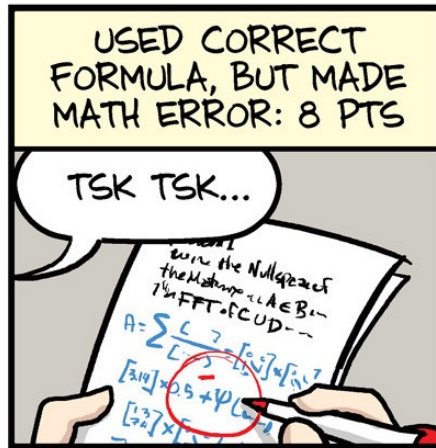
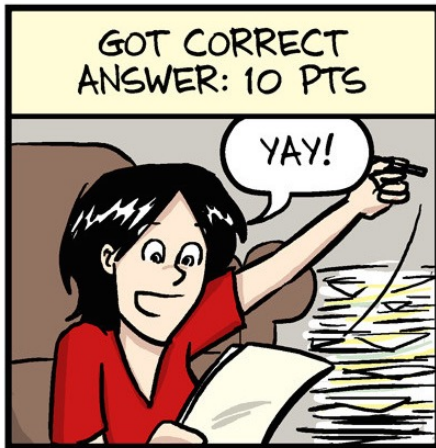
In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.

GRADING RUBRIC

PROBLEM 1 (TOTAL POINTS: 10)



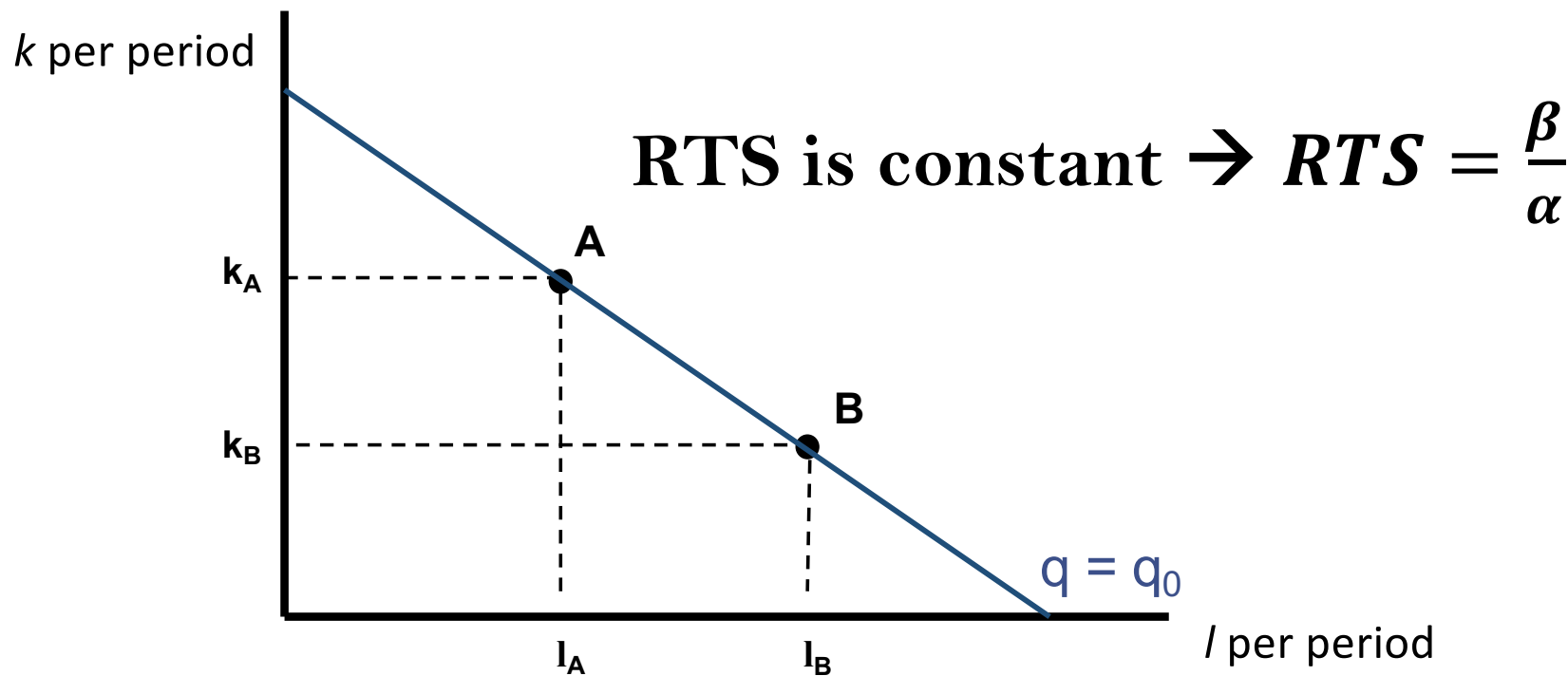
WWW.PHDCOMICS.COM

Activity 5: Solution

Graph the **isoquants** of the following production functions:

a) Linear production function

$$q = f(k, l) = \alpha k + \beta l$$

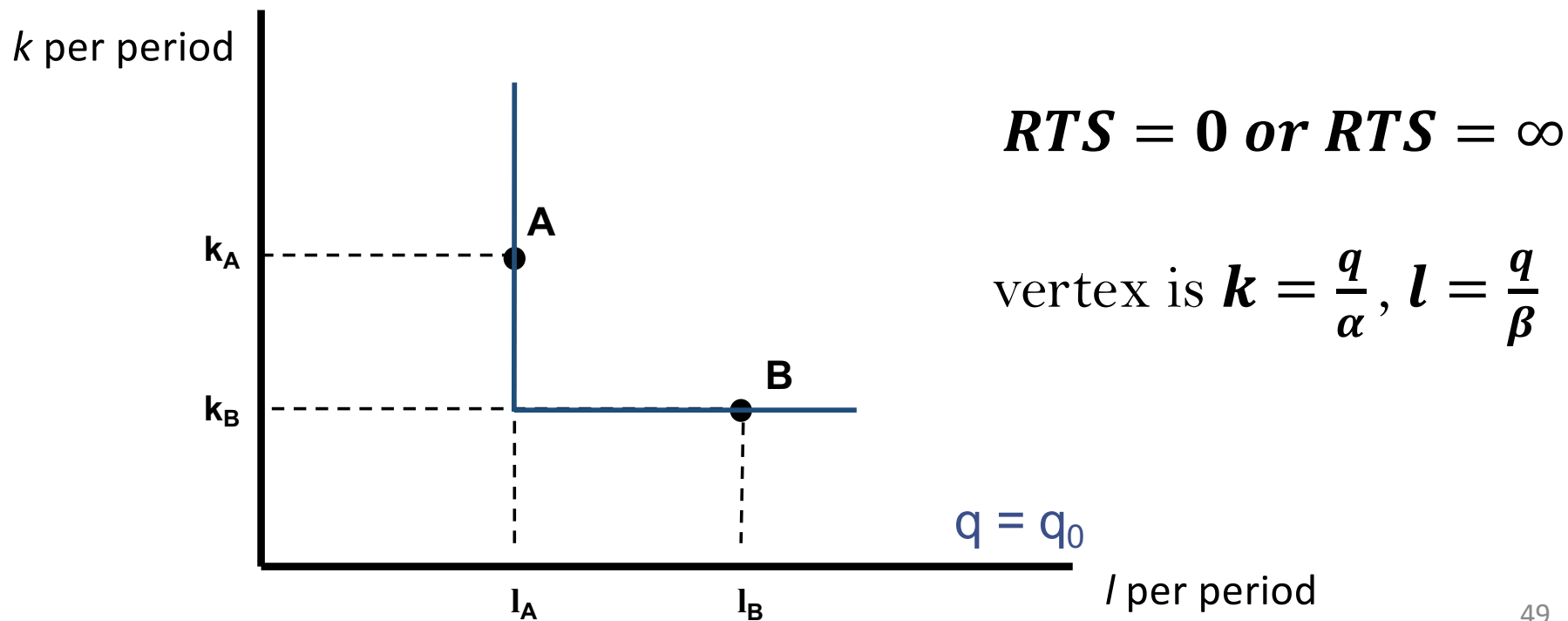


Activity 5: Solution

Graph the **isoquants** of the following production functions:

b) Fixed proportions

$$q = f(k, l) = \text{Min} \{ \alpha k, \beta l \}$$



Activity 5: Solution

Graph the **isoquants** of the following production functions:

c) Cobb-Douglas

$$q = f(k, l) = Ak^{\alpha}l^{\beta}$$

